PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Automated Lunar Creation Detection Using Deep Learning-Based Object Detection (YOLOv8 Framework)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Tanishjot Kaur

2210992443

Kanisha Jain

2210991737

Supervised By:

Dr. Monika Aggarwal

Mentor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Automated Lunar Crater Detection Using Deep Learning-Based Object Detection (YOLOv8 Framework)"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision **of Dr. Monika Aggarwal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Tanishjot Kaur (2210992443)**

Date : 1 May 2026 **Kanisha Jain (2210991737)**

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr . Monika Aggarwal

Mentor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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ABSTRACT

Mapping craters on the Moon is a core part of planetary science. By studying the size and number of craters in an area, researchers can figure out the age of different lunar surfaces and track the history of asteroid impacts. Satellites like the Lunar Reconnaissance Orbiter currently send back millions of highly detailed pictures of the Moon. But manually having humans draw boxes around every single crater takes too much time. Early computer programs tried to automate this by searching for circle shapes and shadows. Unfortunately, those older methods usually failed because the Moon has harsh lighting and many overlapping craters that confuse basic algorithms.

In this project, we created a fully automated system to spot lunar craters using YOLOv8, a modern deep learning model. We trained our AI on a massive collection of black-and-white satellite images. Before giving the images to the model, we used contrast-enhancement techniques to help the AI see through pitch-black shadows. After 50 rounds of training, the model became highly skilled at identifying craters of all sizes. It reached a Mean Average Precision (mAP) of 0.924, proving it is extremely reliable. This research shows that fast, modern object detection can map the Moon's surface automatically, which will be essential for finding safe landing spots for upcoming space missions.

1. Introduction

1.1. Background and Motivation

When you look at the Moon, the craters are the first thing you notice. These marks have been forming for billions of years as meteors and asteroids crashed into the surface. Since the Moon does not have wind, rain, or an active atmosphere, these impact sites never wash away. They stay frozen in time, acting as a historical record of our solar system. Scientists rely heavily on craters using a technique called "crater counting" to estimate the age of the ground. Furthermore, if we want to send rovers or astronauts back to the Moon during the upcoming Artemis missions, we need perfectly accurate maps so they don't crash into a hidden ditch or a steep rocky edge.

1.2. The Problem Statement

For decades, mapping these impacts was a completely manual job. Highly trained geologists would sit at their desks, look at satellite photos, and draw circles around every crater they could spot. Humans are great at doing this, but the process is incredibly slow and boring. Because the Moon has no atmosphere to scatter the sun's light, the shadows are pitch black. Old craters often get hit by newer meteors, creating messy, overlapping shapes that completely break standard computer vision rules.

1.3. Project Objectives

- To develop an automated deep learning pipeline capable of identifying and localizing lunar craters in high-resolution satellite imagery.
- To implement the YOLOv8 object detection framework to ensure both high accuracy and real-time processing speeds.
- To evaluate the model's performance using standard computer vision metrics such as Precision, Recall, and Mean Average Precision (mAP).

2. Methodology

2.1. Dataset Preparation

We used the LU3M6TGT Lunar Crater Dataset for our training. This is a massive folder of pictures taken by lunar orbiters. We used grayscale images because planetary cameras focus on capturing sharp textures rather than color. The dataset features totally different types of terrain, including the smooth, dark volcanic plains known as the "maria," as well as the bright, rugged, heavily cratered "highlands."

2.2. Cleaning Up the Images

Raw images are rarely ready for AI right out of the box. We scaled every single image to exactly 640x640 pixels to match the YOLOv8 input requirements. To fix pitch-black shadows, we used a filter called Contrast Limited Adaptive Histogram Equalization (CLAHE). It looks at small chunks of the image and adjusts the brightness locally.

2.3. Data Augmentation

To prevent the AI from just memorizing images, we used data augmentation:

- Flipping and Rotating: We randomly spun the images and flipped them upside down.
- Mosaic Augmentation: Built into YOLO, this takes four totally different pictures, shrinks them down, and stitches them into a single square to help the AI handle scale differences.

2.4. Why YOLOv8 Works So Well

YOLOv8 served as the main engine for our project. The biggest advantage of YOLOv8 is that it is "Anchor-Free." Older models used to drop hundreds of invisible, fixed-size boxes (anchors) onto an image. Because lunar craters can be 10 meters wide or 100 kilometers wide, fixed boxes rarely fit right. YOLOv8 completely removes that step, predicting the exact center point of the crater and measuring how far the edges are from that spot.

3. Tools and Technologies

3.1. Software Requirements

- **Python 3.x:** The core programming language used for scripting and data manipulation.
- **Ultralytics YOLOv8:** The primary deep learning framework used for object detection.
- **PyTorch:** The underlying open-source machine learning library used by YOLO to compute tensors and neural network weights.
- **OpenCV:** Used for image preprocessing, specifically applying the CLAHE filter and resizing images.

3.2. Hardware Requirements

- **Cloud Computing:** Google Colab Environment.
- **GPU:** NVIDIA T4 GPU (16GB VRAM) was strictly required to handle the massive matrix multiplications required during 50 epochs of training.

4. Implementation

4.1. Environment Setup

The implementation began by setting up a Google Colab notebook and installing the Ultralytics library. The LU3M6TGT dataset was mounted into the environment, and a custom data.yaml file was created to point the YOLO framework toward our specific training and validation image directories.

4.2. Model Training Process

We initialized the YOLOv8 model using pre-trained weights to speed up learning. The model was trained using the Stochastic Gradient Descent (SGD) optimizer. We set our batch size to 16 and commanded the model to run for 50 "epochs," meaning the AI reviewed our entire image dataset 50 separate times, adjusting its internal weights to minimize the bounding box error loss on every pass.

5. Major Findings/Outcomes/Output/Results

5.1. Training Metrics

The Box Loss dropped drastically in just the first 10 epochs. By epoch 50, the model hit a Precision of 0.872 (how often the AI is correct when it claims it found a crater) and a Recall of 0.851 (how many of the real craters in the image the AI managed to find). We achieved a Mean Average Precision (mAP@0.5) of 0.924, which is considered a highly successful and reliable model in computer vision.

5.2. Performance Comparison

To put our results into perspective, we put our numbers side-by-side with previous famous crater detection studies.

- Cohen et al. (Early CNN): ~0.890 (F1-Score)
- Silburt et al. (U-Net): 0.920 Recall
- Chen et al. (Deep Learning): 0.829 Precision, 0.793 Recall
- **Our Proposed Model (YOLOv8):** 0.872 Precision, 0.851 Recall, 0.924 mAP. Our YOLOv8 setup balances both Precision and Recall brilliantly, resulting in a top-tier mAP score that outperforms older architectures.

5.3. Visual Testing

When fed a batch of unseen images, the AI successfully drew precise bounding boxes around craters. It handled heavy clusters perfectly, separating overlapping impacts. Furthermore, it spotted "fakes" by ignoring long moon cracks and topographical ridges that cast shadows similar to a crater wall.

6. Conclusion and Future Scope

6.1. Conclusion

We successfully built and trained a deep learning system that can automatically hunt down and map craters on the Moon using the YOLOv8 framework. By cleaning up our image data and letting the AI study the patterns of the lunar surface, we hit a very high accuracy score of 0.924. This proves that modern AI is more than tough enough to handle the weird, harsh visual conditions of outer space, performing in a fraction of a second what would take humans hours to do manually.

6.2. Future Scope

- **Deploying on Rovers:** Because our YOLOv8 model is so fast and lightweight, we could theoretically copy this code onto a small computer like a Raspberry Pi and strap it to a lunar rover for real-time hazard avoidance.
- **Mapping Other Planets:** We plan to use Transfer Learning to show the AI pictures of Mars and Mercury so it can map out those planets just as easily.
- **Adding 3D Depth:** We want to combine our standard satellite photos with laser-altimeter data (DEM) so the AI can not only find the crater but immediately calculate exactly how deep the hole is.

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